

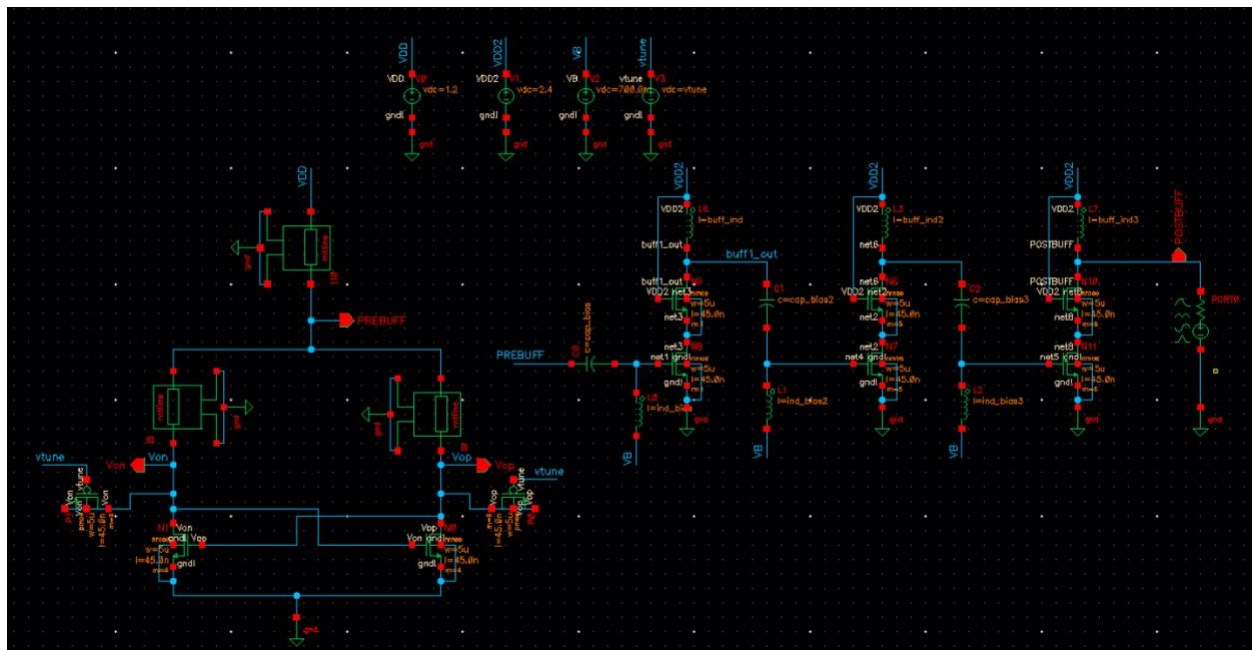
110 GHz Push-Push VCO with 3-stage Buffer Amplifier Chain

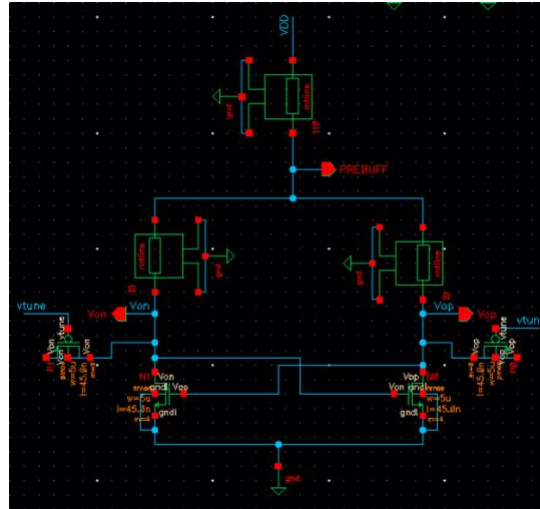
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Project Overview:

This project showcases a 55GHz dual supply rail push-push oscillator that utilizes the transistor parasitic capacitances and transmission lines to attain a 55GHz fundamental frequency and 110GHz second harmonic. The architecture uses a cross-coupled configuration to introduce negative resistance and sustain oscillation that would have been otherwise lost to resistive losses. Varactors are added to the fundamental harmonic to achieve 10% frequency tuning. The 3-stage cascode buffer amplifier improves the output power from -4dBm to +7.3dBm. Phase noise was calculated to be -68 dBc/Hz. Power consumption of the full circuit resulted in 150mW and a PAE of 4%

Final Stage:





The transmission lines, labeled mtline act as inductors. The symmetric tlines in the oscillator tank are the inductors used for oscillation while the tline connected to VDD is used to direct the signal into the prebuff node.



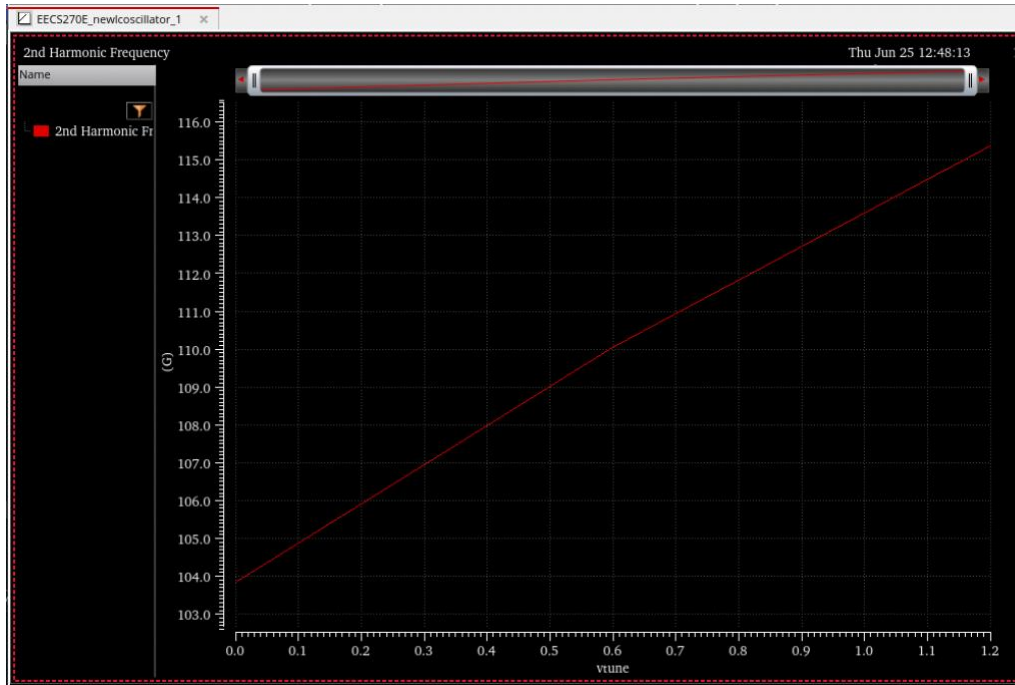


Figure: Second Harmonic Tuning Range (104G – 116G)

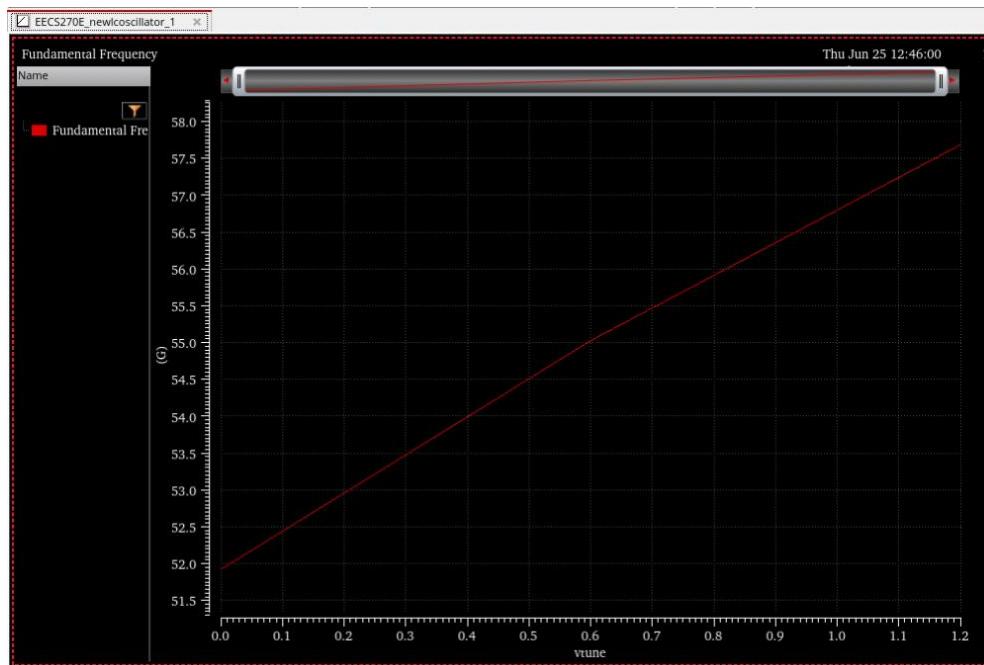
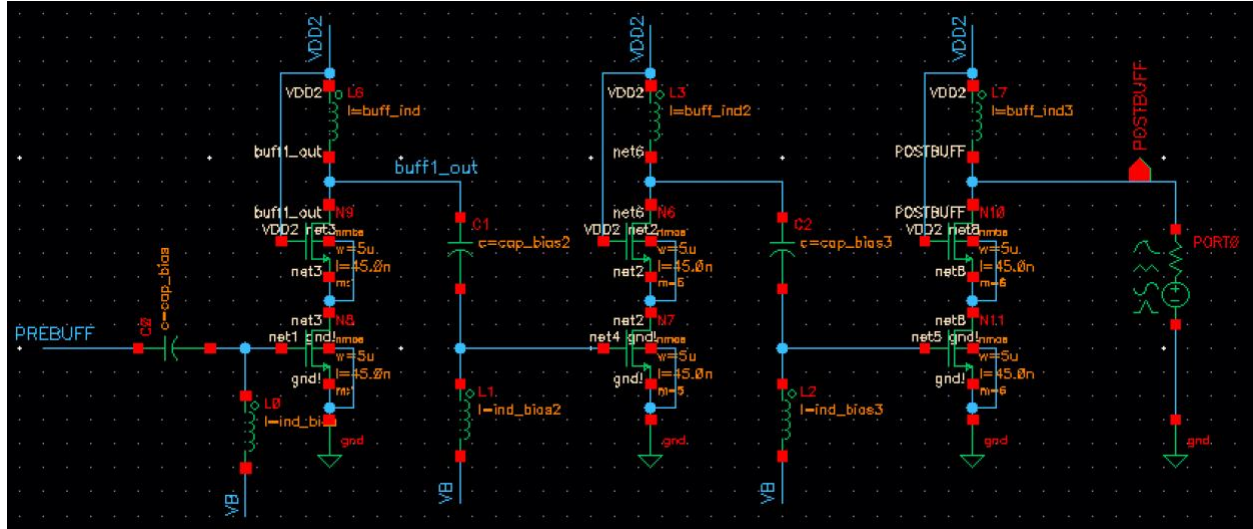


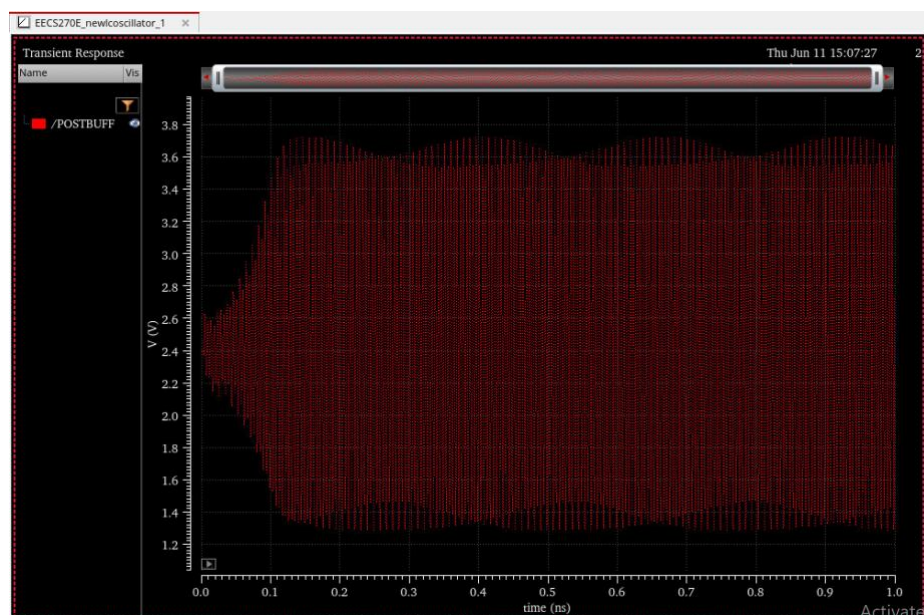
Figure: Fundamental Harmonic Tuning Range (51.9G – 57.6G)

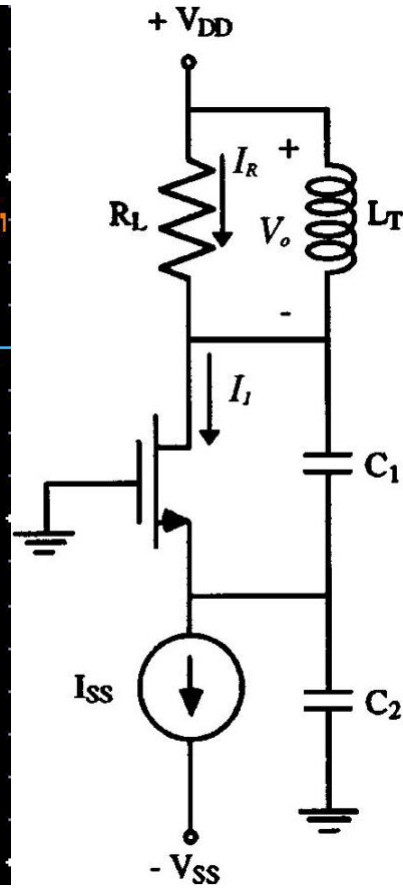
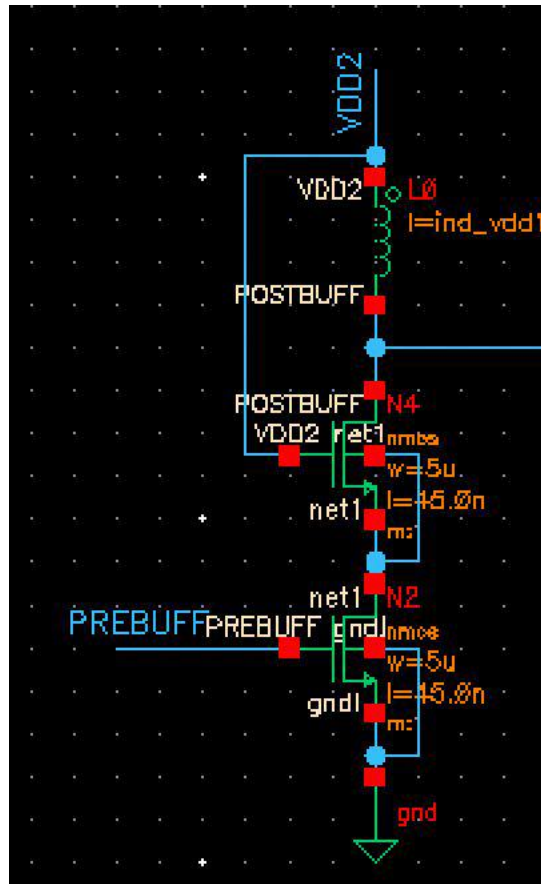
Attached to the fundamental harmonic nodes are pmos varactors (source and drain tied together) with a width of 5um and multiplier of 9 resulting in a total width of 45um. The fingers parameter was not used because after multiple simulations, I realized that they

were not making a difference and the parasitics remained the same. With a sweep of the tuning voltage from 0v – 1.2v, the circuit achieved a range from 51.9G – 57.6G.

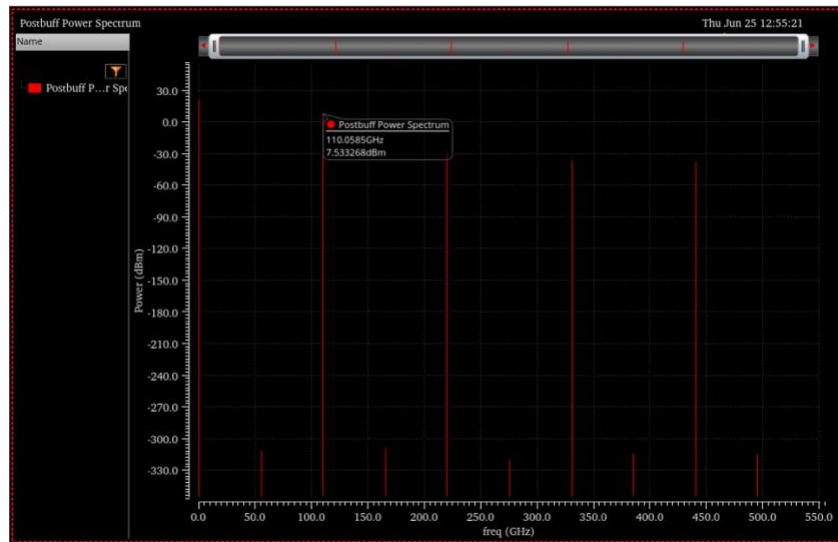


Three sets of cascode amplifiers are used to buffer the oscillator output and drive it to the next stage. A cascode configuration was chosen to limit the miller capacitance. Each stage uses has an inductor close to VDD2 (2.4v) to prevent the signal from flowing into AC ground. Between each stage including the input to the first stage is an RF choke to rebias the driver mosfet of each amplifier.





Without the RF choke, the VDD2 bias is set as the operating point and forces the driving transistor into triode. The interesting issue that arises without the RF choke is that the amplifier stage turns into a Colpitts oscillator due to the transistor parasitics and therefore the output captures both the frequency of the VCO and the Colpitts resulting in the above waveform.

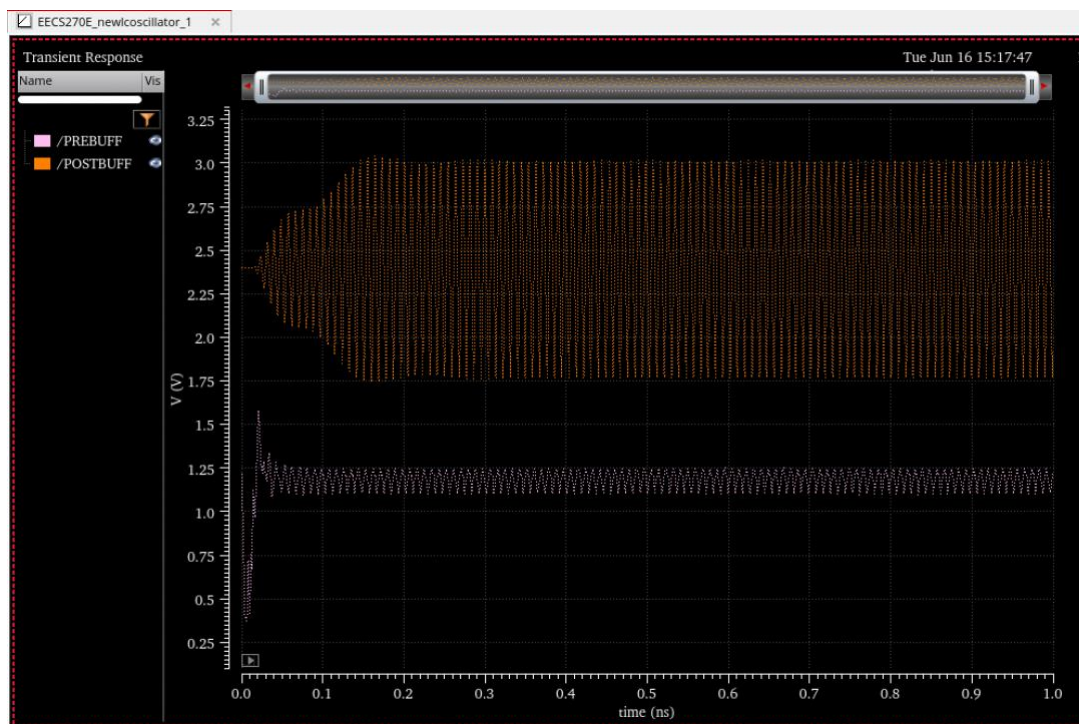


Connected to the last stage is a 50 Ohm port that allows for power extraction. Without a matching network, the output power settles at +7.3dBm after all three amplifier stages. Without the three amplifiers, the output power was around -4dBm.

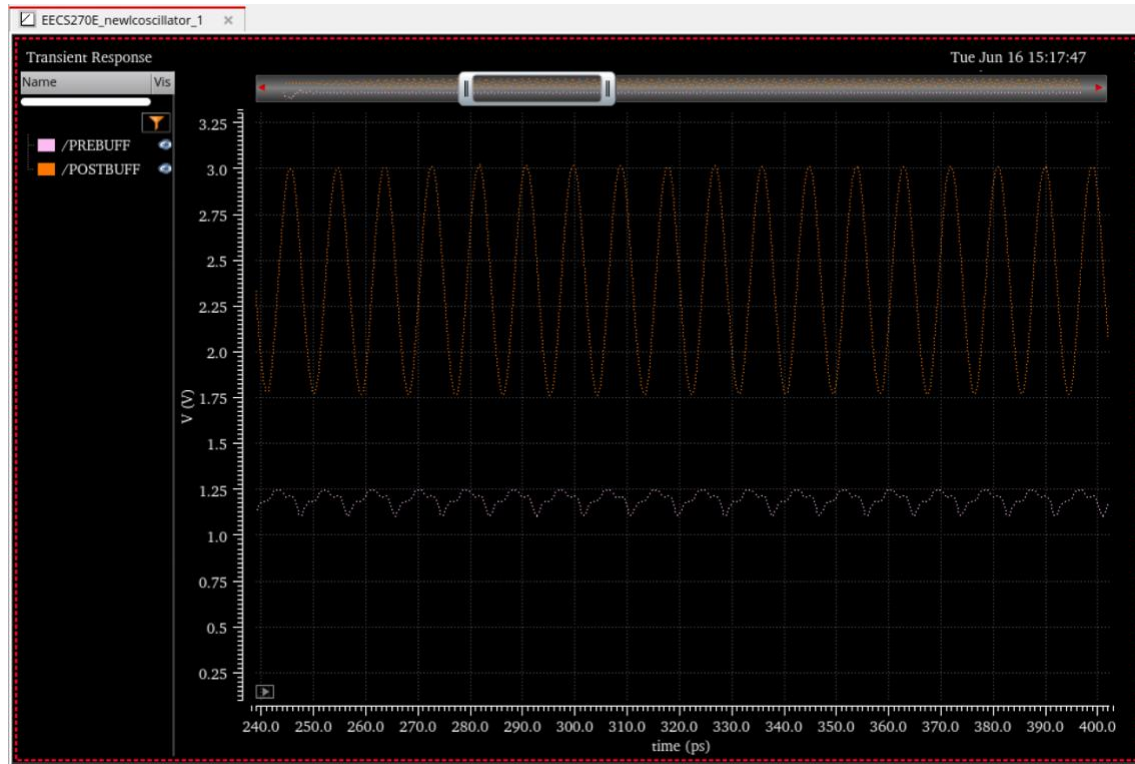
Design Variables		
len_drain	270u	
len_vdd	70.56u	
buff_ind	217.1p	
cap_bias	21.56f	
ind_bias	64.97p	
buff_ind2	87.4p	
cap_bias2	3.427f	
ind_bias2	52.86p	
buff_ind3	67.5p	
cap_bias3	77.4f	
ind_bias3	52.8p	
vtune	0.6	

Results:

Transient Simulations:



Prebuff is the output of the oscillator when loaded with the amplifier chain and postbuff is the waveform after the amplifier chain.



Power consumption:

power consumption - oscillator	expr	(average(I(T"/I10/in1")) * 1.2)	12.37m	☒	☐
total power consumption	expr	((mag(IDC("/V1/PLUS")) * 2.4) + (mag(IDC("/V0/PLUS")) * 1.2))	150.3m	☒	☐

As seen, the power consumption is high. If changes were to be made, I think changing the type of amplifier configuration will decrease the power consumption. I can use a common source configuration, use a one supply rail and limit millers' effect through neutralization.

PAE:

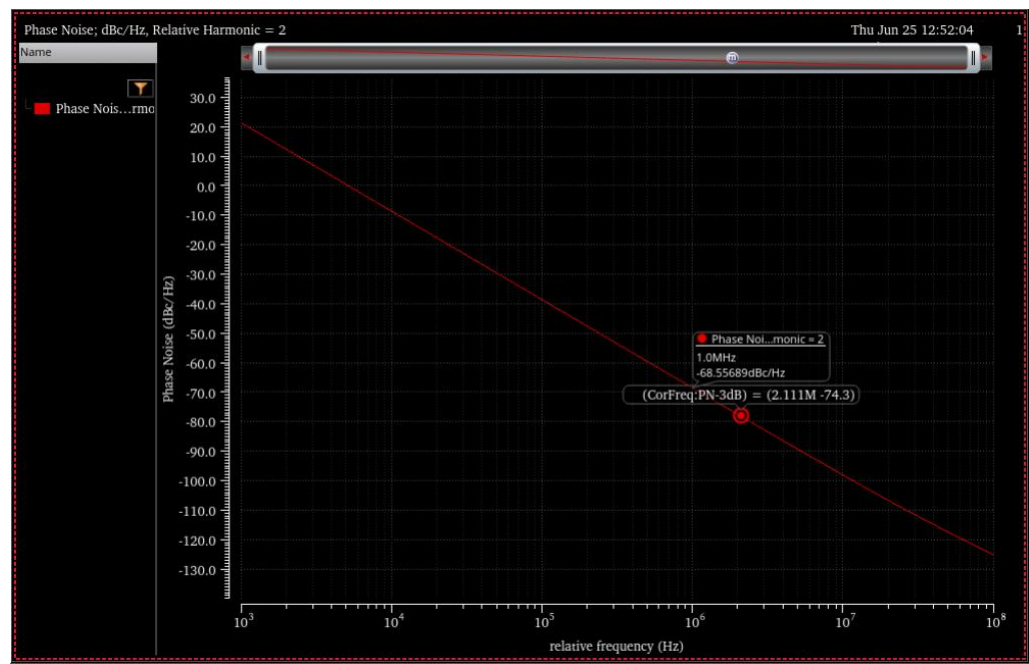
$$\text{PAE}(\%) = \frac{P_{\text{RF,out}} - P_{\text{RF,in}}}{P_{\text{DC,buffer}}} \times 100\%$$

$$\text{PAE} = \frac{5.666 \text{ mW} - 0.119 \text{ mW}}{137.93 \text{ mW}} \times 100\%$$

$$\text{PAE} = \frac{5.547 \text{ mW}}{137.93 \text{ mW}} \times 100\% \approx \mathbf{4.02\%}$$

PAE is 4%.

Phase Noise:

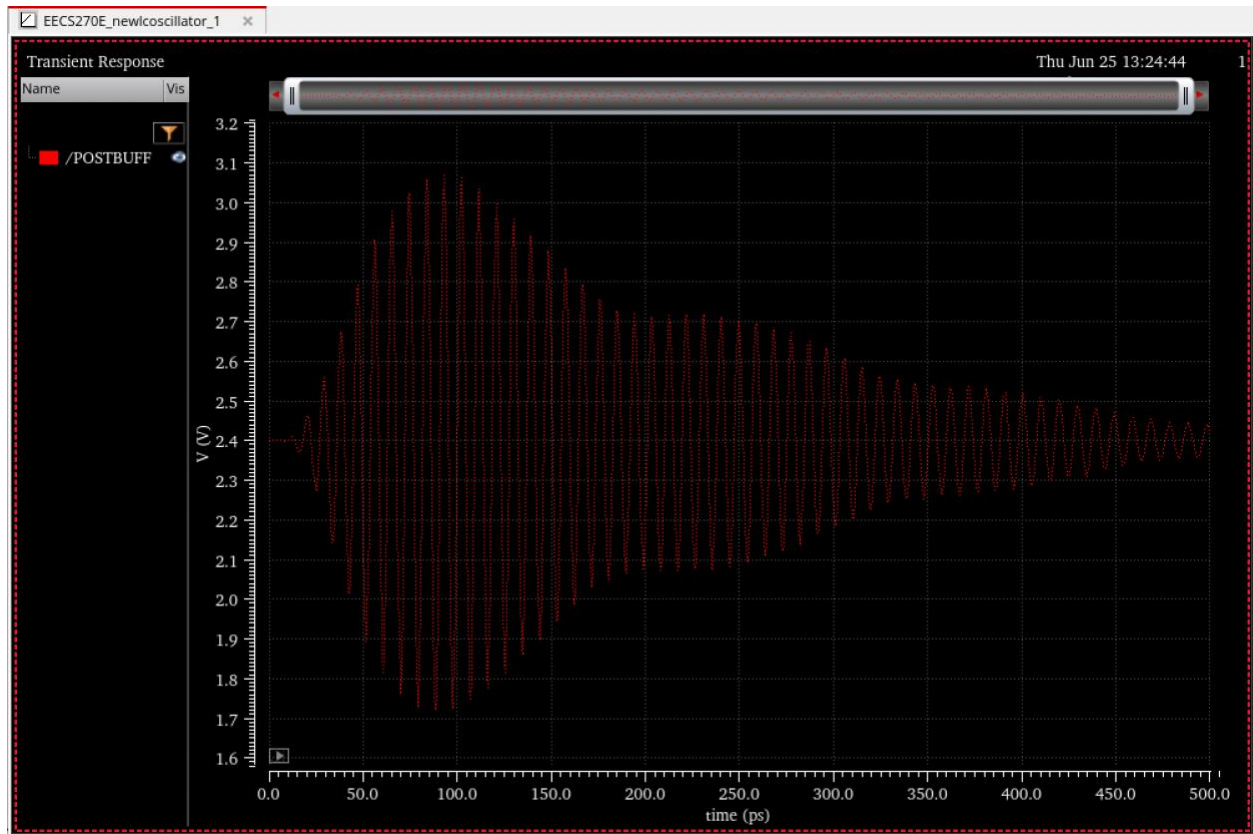


The phase noise sits at -68 dBc/Hz.

Name	Type	Details	Value	Plot	Save	Spec
Filter	Filter	Filter	Filter			
vop	signal	/Vop		☐	☐	
von	signal	/Von		☐	☐	
prebuff	signal	/PREBUFF		☐	☐	
buff1_out	signal	/buff1_out		☐	☐	
postbuff	signal	/POSTBUFF		☐	☐	
Phase Noise; dBc/Hz, Relative Harmonic = 2	expr	pn("pnoise")		☐	☐	
power consumption	expr	(average(I("I10/in1")) * 1.2)	12.37m	☒	☐	
Fundamental Frequency	expr	harmonic(xval(getData("Vop" ?result "hb_fd")) '1)	55.03G	☒	☐	
2nd Harmonic Frequency	expr	harmonic(xval(getData("Vop" ?result "hb_fd")) '2)	110.1G	☒	☐	
Buff 1 Output Power	expr	dbm(harmonic(pvr("hb "/buff1_out" 0 50.0) 2))		☐	☐	
Prebuff Peak to Peak	expr	(ymax(VT("/PREBUFF")) - ymin(VT("/PREBUFF")))		☐	☐	
Prebuff Output Power	expr	dbm(harmonic(pvr("hb "/PREBUFF" 0 50) 2))		☐	☐	
Postbuff Output Power	expr	dbm(harmonic(pvr("hb "/POSTBUFF" 0 50) 2))	7.533	☒	☐	
Postbuff Power Spectrum	expr	dbm(pvr("hb "/POSTBUFF" 0 50))		☒	☐	
Vop Output Power	expr	dbm(pvr("hb "/Vop" 0 50))		☐	☐	
Von Output Power	expr	dbm(harmonic(pvr("hb "/Von" 0 50) 2))		☐	☐	
2nd Har Power Spectrum	expr	dbm(pvr("hb "/PREBUFF" 0 50))		☐	☐	
S11 dB20	expr	db(spm("sp 1 1))		☐	☐	
Z11 reOhm	expr	real(zpm("sp 1 1))		☐	☐	
Z11 imOhm	expr	imag(zpm("sp 1 1))		☐	☐	
Corner Freq	expr	"2.111e+06;-7.43e+01"		☐	☐	

Stability:

Stability is still an issue that hasn't been fixed yet. Because of oscillations that arise from the parasitics in the buffer stage transistors, the resulting waveform when the VCO is turned off is shown in the waveform below.



The signal's voltage swing is significant and slowly decreases but the time that it takes is still too long. Potential fixes are to add passive components to dissipate the extra power.